

Weather Impact Analysis System Report/Summary

A Python case study that analyzes daily weather reports from two farms — Kaduna and Benue — to determine whether rainfall and temperature conditions favor crop growth, and whether the farm manager needs to take action.

Goal

- Compare recorded rainfall with the minimum rainfall requirement.
- Check whether temperature is within the safe limit.
- Determine whether growing conditions are favorable.
- Recommend management action based on the result.

Data

Field	Kaduna	Benue
Crop	Maize	Rice
Temperature	31°C	38°C
Updated rainfall	26 mm	20 mm
Minimum rainfall required	20 mm	25 mm
Maximum safe temperature	35°C	35°C
Irrigation available	Yes	No

Key Findings

- **Kaduna** — rainfall requirement met, temperature within safe limit, irrigation available → **favorable conditions, no action required**.
- **Benue** — rainfall below requirement, temperature above safe limit, no irrigation → **unfavorable conditions, action required**.
- Of the two, Benue is the priority farm for an extension officer to visit.

Tools

Python · Jupyter Notebook

How to Run

1. Clone this repository.
2. Open `Project/Weather_Impact_Analysis_System.ipynb` in Jupyter Notebook.
3. Run all cells to see the full analysis and generated report.

Repository Structure

- **Project/** – Python notebook and project files.
- **Presentation/** – Presentation slides.
- **Report/** – Full project report.

Author

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